# Supplementary Information for: How do extreme weather events affect fishing operations? Knowledge engineering a risk analysis with Swedish Baltic Sea fishers

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## Supplementary Information C: Conditional probability tables

## Contents

**Network structure** **3**

**Algebraic methods for Ranked nodes** **5**

**Decision nodes** **6**

Extreme weather events (node: extreme_event) . . . 6
Fishing style (and acronyms) . . . 6
Aware of the event . . . 6

**Chance nodes - Operational** **7**

Strategy to_change . . . 7
Additional mitigation . . . 8
Gear . . . 9
Catchability . . . 10
Catch condition . . . 11
Personal safety . . . 12
Damage . . . 13
Target . . . 14
Go fishing . . . 15

**Chance nodes - User** **16**
Stress . . . 16
Health . . . 17
Satisfaction . . . 18
Substitution capacity . . . 19
Non monetary_value . . . 20

**Chance nodes - Economic** **21**
Costs . . . 21
Monetary loss . . . 22
Credit . . . 23
Job mobility . . . 24
Economic buffers . . . 25

**Chance nodes - Individual** **27**
Identity . . . 27
Sense of_home . . . 28
Outdoor benefit . . . 29
Individual importance . . . 30

**Chance nodes - Societal** **32**
Knowledge transmission . . . 32
Benefit local_community . . . 33
Traditional food . . . 34
Benefit personal_community . . . 35
Societal importance . . . 36

**Utilities - Risk** **37**
Individual risk . . . 38
Societal risk . . . 39
Economic risk . . . 40

**References** **41**

## Network structure

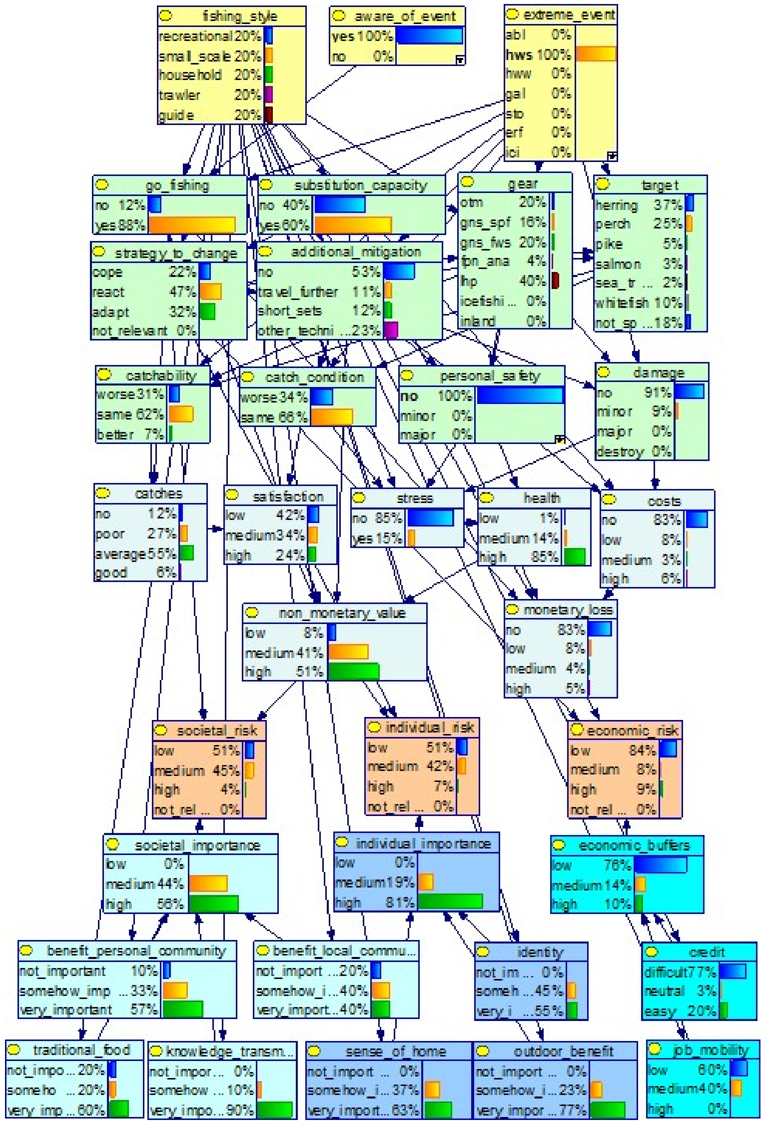


Figure C.1: Bayesian Network representing risks to Swedish fisheries posed by Marine Extreme Weather events (abl algal bloom, hws summer heatwave, hww winter heatwave, gal gale, sto storm, erf extreme railfall, ici icing) effects on fishing operations, as identified at the conclusion of the third cycle of stakeholder (fishers) dialogues. The probability distributions shown in the boxes represents the example of how summer heatwave affects fishing operations across all fishing styles. Conditional probability distribution details and terminology are reported in Supp. Info. C. Colors are associated to node types and group Yellow: decision nodes; lime green: chanche nodes – operational group; grey: chanche nodes - user group; navy blue: chance nodes - individual group; light blue: chance nodes - societal group; cyan: chance nodes - economic group; orange: risk nodes

# Algebraic methods for Ranked nodes

Ranked nodes have order-preserving meaning states (e.g.: low/medium/high) that can be mapped to a continuous numerical scale bounded to [0,1], assuming equal intervals (Norman E. Fenton and Neil 2019). For instance, in a node having three states (low, medium, high), the "low" state takes the interval [0-0.33). The probability distribution of a ranked node in the numerical scale is described as $TNormal(\ \mu\ ,\sigma^2,0,1)$. The use of a numerical scale allows us to use linear algebra to perform operations among variables, a technique that is particularly useful to build a CPT from a reduced number of questions (N. E. Fenton, Neil, and Caballero 2007). When a child has multiple parents with multiple states (binary or more), the distribution of a child node can be calculated by implementing algorithms based on the state, direction (positive or negative) and weight of their parents. The algorithms most commonly applied to ranked nodes gives equal (Eq. B1) or unequal contribution to states. Weighted max (Eq. B2) assigns more contribution to large states, while Weighted min (Eq. B3) assigns more contribution to small states.

$$\mu = \frac{\sum_{i=1}^{n} w_i X_i}{\sum_{i=1}^{n} w_i}$$

[Eq.C1]

$$\mu = min[\frac{w_i X_i + \sum_{i \neq j}^{n} X_j}{w_i + (n-1)}]$$

[Eq.C2]

$$\mu = max[\frac{w_i X_i + \sum_{i \neq j}^{n} X_j}{w_i + (n-1)}]$$

[Eq.C3]

# Decision nodes

## Extreme weather events (node: extreme_event)

The node *extreme_event* is used to wrap all the extreme weather events we are considering.

*Levels (and acronyms)*

- **Storms** (sto). Strong wind events, with intensity above 24.5 m/s. Season: summer.
- **Gales** (gal), or moderately strong wind events, with intensity between 15 m/s and 24.5 m/s;
- **Extreme precipitation** (erf) – rainstorm (summer), or deviation from average precipitation regime;
- **Algal bloom** (abl): intense algal bloom;
- **Summer heatwaves** (hws): prolonged discrete anomalously warm water event occurring in summer (Hobday et al., 2016);
- **Winter heatwaves** (hww): prolonged discrete anomalously warm water event occurring in winter (Hobday et al., 2016);
- **Icing** (ici): strong wind events associated to ice, which then forms icing pile up on gears and boats.

## Fishing style (and acronyms)

The node *fishing_style* is used to wrap all the fishing styles and seasons.

*Levels (and acronyms)*

- **Recreational fisher (recreational - rcf)**: fishers without a commercial fishing license, deploying hand-held gears for recreational purposes;
- **Fishing guides (guides - fgu)**: commercial actors who provide recreational fishers with fishing experience by sharing expert and local knowledge, giving access to angling concession and offering boat chartering ;
- **Small-scale coastal fishing (small_scale - ssf)**, fishers operating small boats (<12 m) deploying species-specific passive gears.
- **Pelagic trawling (trawlers - trw)**, fishers operating medium to large size vessels (24-60 m) deploying pelagic trawl nets;
- **Household fishing (household - hou)**, fishing by residents of the Swedish archipelago, using coastal-type passive gears, without commercial licences and without legal access to markets.

## Aware of the event

The node *aware_of_event* controls the fisherman awareness of the event occurrence. In the case when fisher is not aware, they always goes out fishing. The fisher can still adopt react or adapt str

*Levels (and acronyms)*

- **yes**: the fisher is aware of the extreme event. Decisions whether to go out fishing or not are taken according to interviews
- **no**: the fisher is not aware of the extreme event. The probability of going out fishing are 100%

# Chance nodes - Operational

## Strategy to_change

**Parents**: fishing_style, extreme_event.

**Levels**: cope, react, adapt, not_relevant.

**Description**: Fishers strategy is coded as adapt, react or cope (Green et al., 2021). Adapt is proactive planning, it implies large changes in fishing strategy (i.e.: changing fishing target and/or gear). React is unplanned response, or small modification to fishing practice (fishing in a diferent location or in different time of the day). Cope is passively accepting consequences, it implies no modifications at all to the fishing strategy.

**Method**: Case C1. We interpreted narrative (especially questions Q 2, 4, 13 and 15) to associate one or more among Adapt, React or Cope to each combination of fisher and MEW. The question chosen to fill the CPT depends on the narrative.

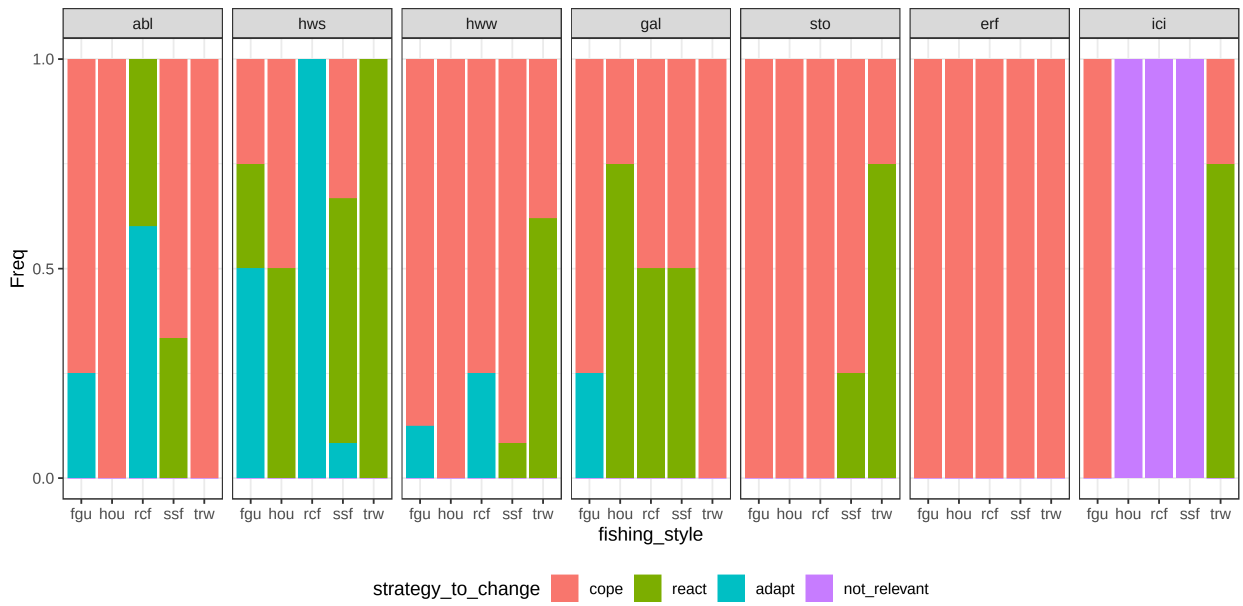


Figure C.2: conditional distribution for strategy_to_change. Columns represents levels of the node extreme_event.

## Additional mitigation

**Parents**: strategy_to_change, fishing_style, extreme_event.

**Levels**: no, travel_further, short_sets, other_technical.

**Description**: Measures implemented in the case when the strategy is React. The different solutions work differently and have different cost: travel further implies fishers change fishing location to seek a sheltered place or to find deeper waters; short sets mean that the fisher perform shorter sessions than average (i.e.: reduced soak time for gillnets); other technical includes strategies that are mentioned sporadically (i.e.: use of ice machines).

**Method**: Case C4. We interpreted narrative to associate one or more solution to each combination of fisher and MEW. In case only one strategy is mentioned, this is assigned probability = 1. When two or more strategies are mentioned, they are given equal probability unless explicitly indicated in the interviews.

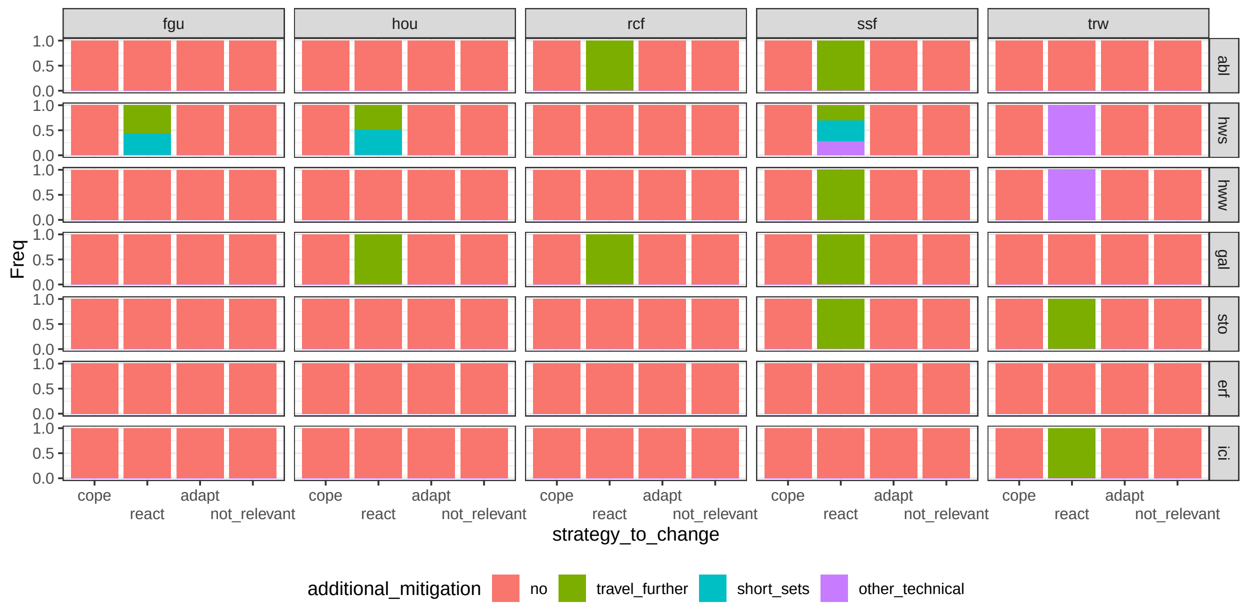


Figure C.3: conditional distribution for additional_mitigation. Columns represents levels of the node fishing_style; rows are levels of the node extreme_event.

## Gear

**Parents**: fishing_style, extreme_event, strategy_to_change.

**Levels**: otm, gns_spf, gns_fws, fpn_ana, lhp, icefishing, inland.

**Description**: Most probable gear used by each fisher. Terminology is mostly inspired by codes used in the European Data Collection Framework (https://dcf.ec.europa.eu/data-calls/definitions-and-terminology/g/gear_en). Otm is pelagic (midwater) otter-trawl; gns_spf is gillnets targeting small pelagic fish; gns_fws is gillnets targeting freshwater species; fpn_ana is traps (or pots) targeting anadromous species; lhp is Handlines and pole-lines (hand-operated); icefishing is fishing with lines by carving holes on iced surfaces of coastal areas and lakes; inland is a general description for any fishing activity poerformed in lakes and rivers.

**Method**: Case C4. In the case when the strategy is cope or react, the CPT reflects literature as described in Annex C. When the strategy is React, it is assigned expert based probability based on interviews interpretation.

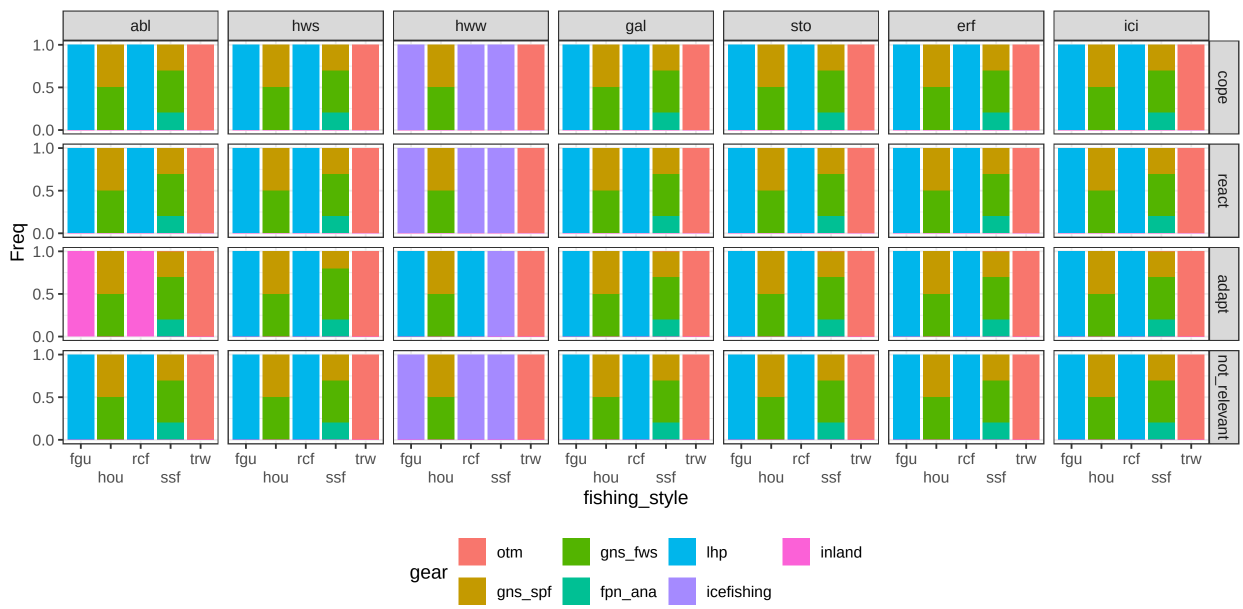


Figure C.4: conditional distribution for gear. Columns represents levels of the node extreme_event; rows are levels of the node strategy_to_change.

## Catchability

**Parents**: target, extreme_event, gear.

**Levels**: worse, same, better.

**Description**: The ability to catch fish depending on gear-fish interactions. For example, a gillnet clogged in algae is more visible and less efficient (less catchability).

**Method**: Case C3 based on question Q 3.


Figure C.5: conditional distribution for catchability. Columns represents levels of the node extreme_event; rows are levels of the node gear.

## Catch condition

**Parents**: target, extreme_event, additional_mitigation.

**Levels**: worse, same.

**Description**: The real or perceived quality of catches in terms of aspect/health status. Does NOT refer to species composition or distribution. There are some "technical solutions" that balance for negative effects of MEW.

**Method**: Case C3 based on question Q5. The CPT is changed to 100% probability of level *same* when appropriate technical solutions (i.e.: travel further, ice machine) are mentioned in the narrative.

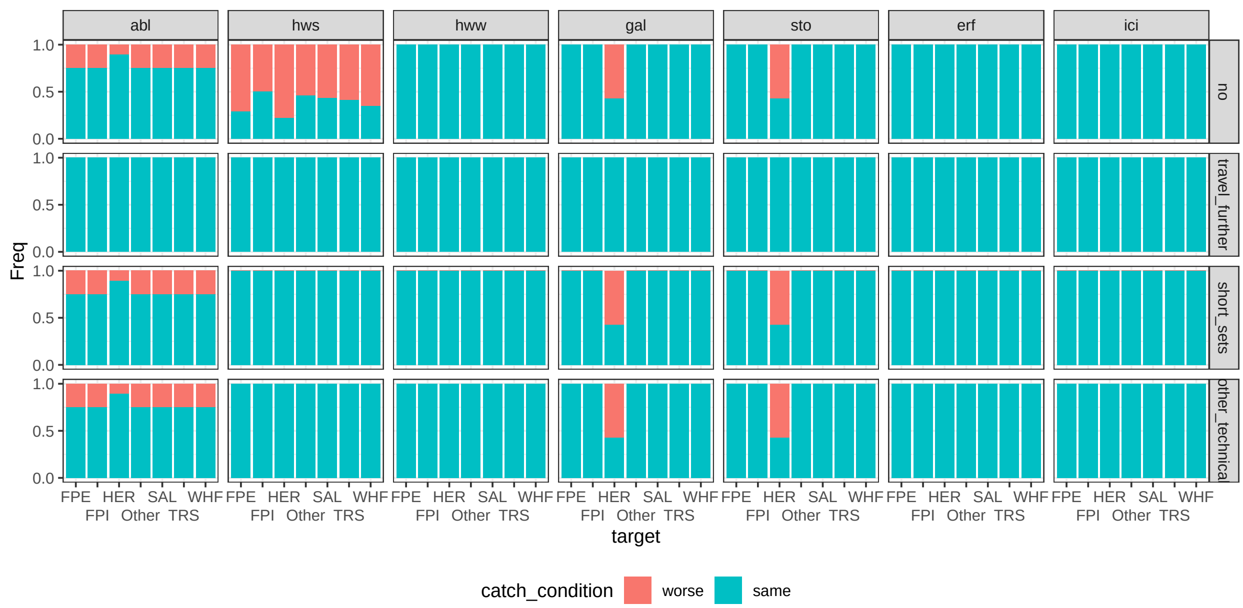


Figure C.6: conditional distribution for catch_condition. Columns represents levels of the node extreme_event; rows are levels of the node additional_mitigation.

## Personal safety

**Parents**: extreme_event, gear, additional_mitigation.

**Levels**: no, minor, major.

**Description**: Any variation to ordinary working conditions that can increase the potential physical damage for operators (i.e.: slipping; a violent storm is life threatening). There are some "technical solutions" that balance for negative effects of MEW.

**Method**: Case C3 based on question Q9. The CPT is changed to 100% probability of level *no* when appropriate technical solutions (i.e.: travel further) are mentioned in the narrative.

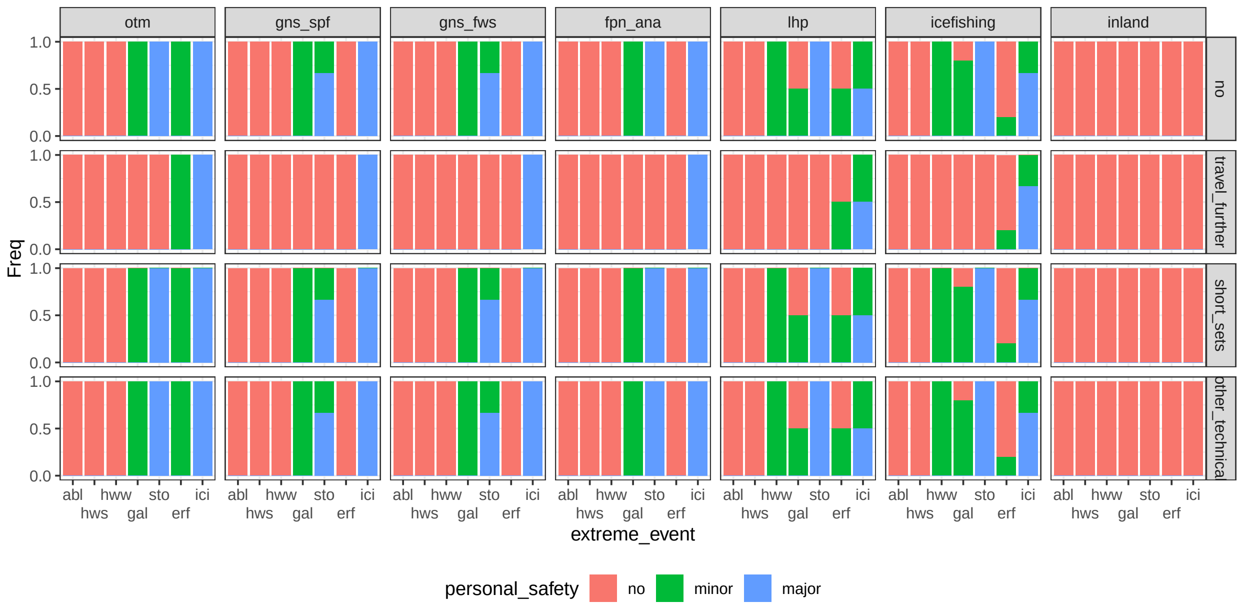


Figure C.7: conditional distribution for personal_safety. Columns represents levels of the node gear; rows are levels of the node additional_mitigation.

## Damage

**Parents**: extreme_event, gear, additional_mitigation.

**Levels**: no, minor, major, destroy.

**Description**: Damage to the fishing gear ranging from minor to completely losing the fishing gear or part of it because it was lost or irreparably damaged. There are some "technical solutions" that balance for negative effects of MEW.

**Method**: Case C3 based on question Q10. The CPT is changed to 100% probability of level *no* when appropriate technical solutions (i.e.: travel further) are mentioned in the narrative.

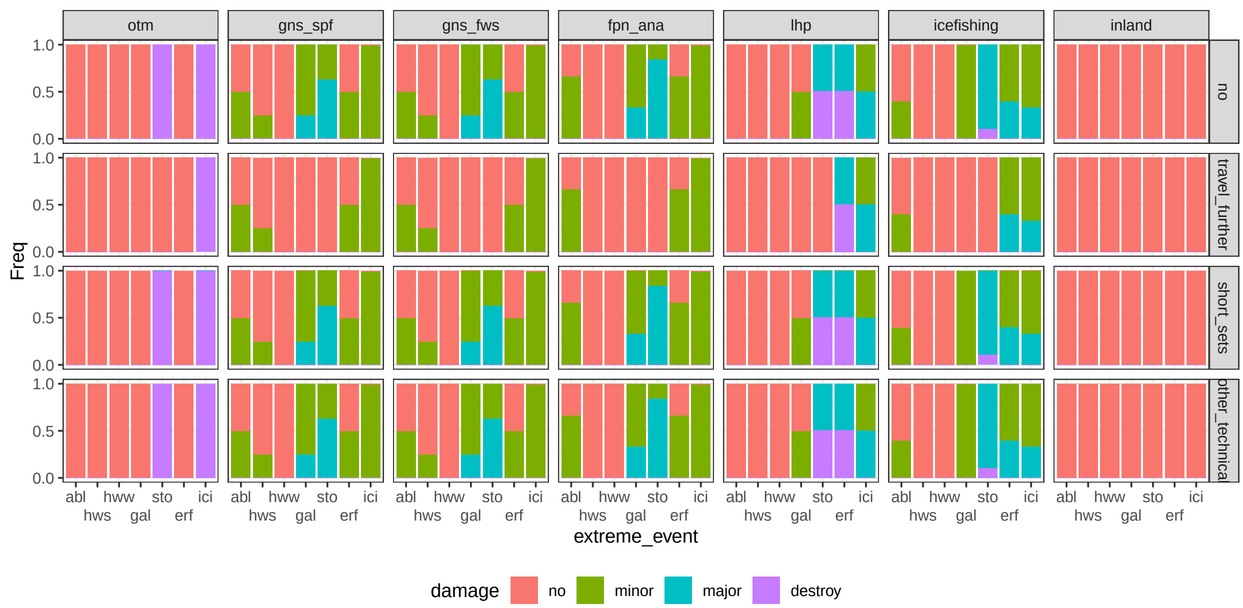


Figure C.8: conditional distribution for damage. Columns represents levels of the node gear; rows are levels of the node additional_mitigation.

## Target

**Parents**: gear, strategy_to_change, extreme_event.

**Levels**: herring, perch, pike, salmon, sea_trout, whitefish, not_specified.

**Description**: The most probable target species associated to each fishing gear. The levels describe the 5 most common species that are relevant across the fishing styles, one aggregation for whitefish (including Common whitefish and vendace), and one category to group any other species. The category not_specified is also assigned to inland and not relevant.

**Method**: Case C4. In the case when the strategy is cope or react, the CPT reflects literature as described in Annex C. When the strategy is React, it is assigned expert based probability based on interviews interpretation.

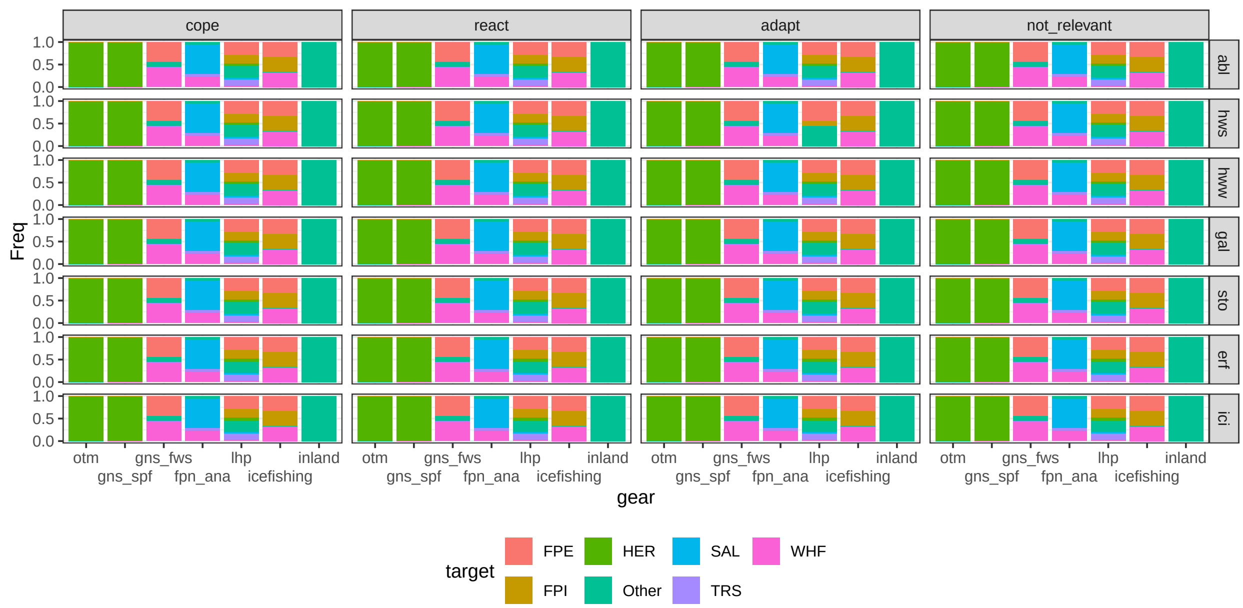


Figure C.9: conditional distribution for target. Columns represents levels of the node strategy_to_change; rows are levels of the node extreme_event.

## Go fishing

**Parents**: extreme_event, fishing_style, strategy_to_change, aware_of_event.

**Levels**: no, yes.

**Description**: The probability to go out fishing.

**Method**: Case C3 based on question Q2 when the level of the parent node strategy is *cope*. The CPT is changed to 100% probability of level *yes* when parent node strategy is *adapt* or *react*.

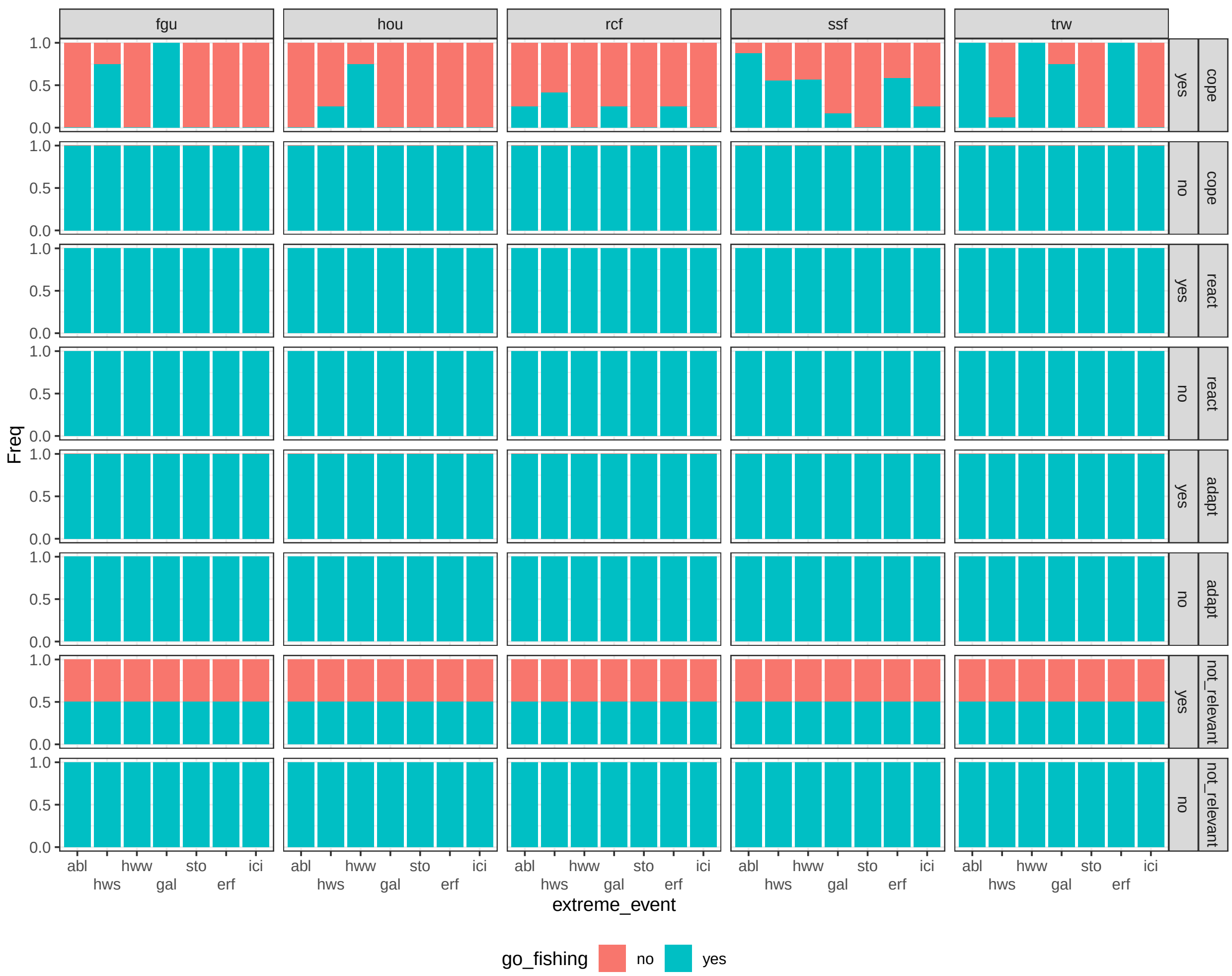


Figure C.10: conditional distribution for go_fishing. Columns represents levels of the node fishing_style; rows are combinations levels of the nodes strategy_to_change and aware_of_event.

# Chance nodes - User

## Stress

**Parents**: personal_safety, damage, catch_condition, fishing_style.

**Levels**: no, yes.

**Description**: Psychological aspect, elements such as anxiety and sense of fatigue (Chatterjee and and Bolar, 2019). The stressful events considered are having damages to the gear, being injuried, being the catch in poor condition.

**Method**: Case C2. Weight based on questions Q21 a, b and c. Algorithm is wmin.


Figure C.11: conditional distribution for stress. Columns represents levels of the node damage; rows are combinations levels of the nodes catch_condition and fishing_style.

## Health

**Parents**: stress, personal_safety.

**Levels**: low, medium, high.

**Description**: Summarizes variables related to physical and mental health. The wmin algorithm implies that all the conditions are necessary to obtain a large value, however the physical injuries are more relevant than stress.

**Method**: Case C2. Weight ratio for the parent *stress* and *personal safety* is 1:3. Algorithm is wmin.

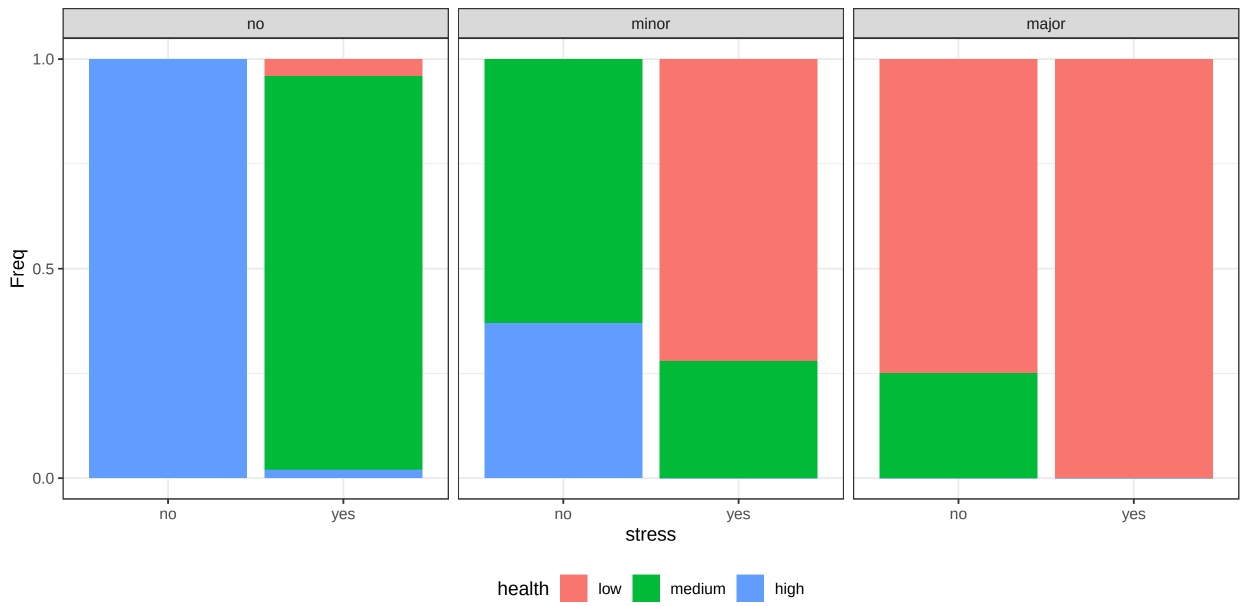


Figure C.12: conditional distribution for health. Columns represents levels of the node personal_safety.

### Satisfaction

**Parents**: catches, catch_condition.

**Levels**: low, medium, high.

**Description**: Summarizes variables related to the quality and amount of catches. The wmin algorithm implies that all the conditions are necessary to obtain a large value.

**Method**: Case C2 with equal weight. Algorithm is wmin.

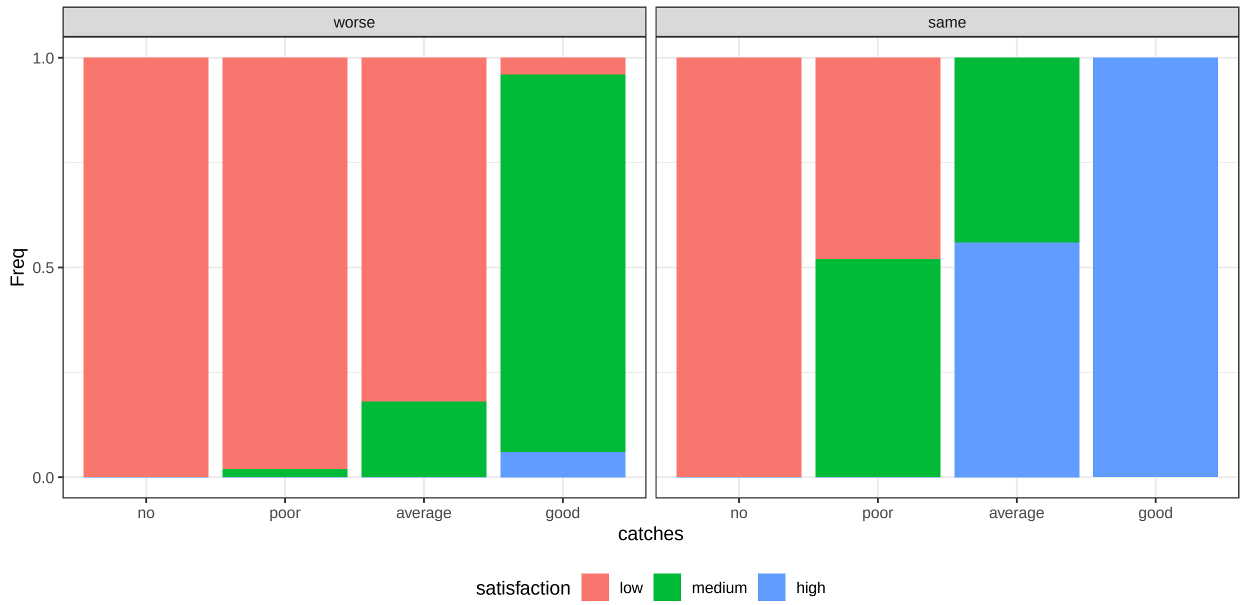


Figure C.13: conditional distribution for satisfaction. Columns represents levels of the node catch_condition.

### Substitution capacity

**Parents**: fishing_style.

**Levels**: no, yes.

**Description**: The impact of a lost day is not determined by the loss itself, but by its recoverability. When value is tied to outcomes, time can be lost and later absorbed. When value is tied to time itself, loss becomes irreversible.

**Method**: Case C1 based on extra question.

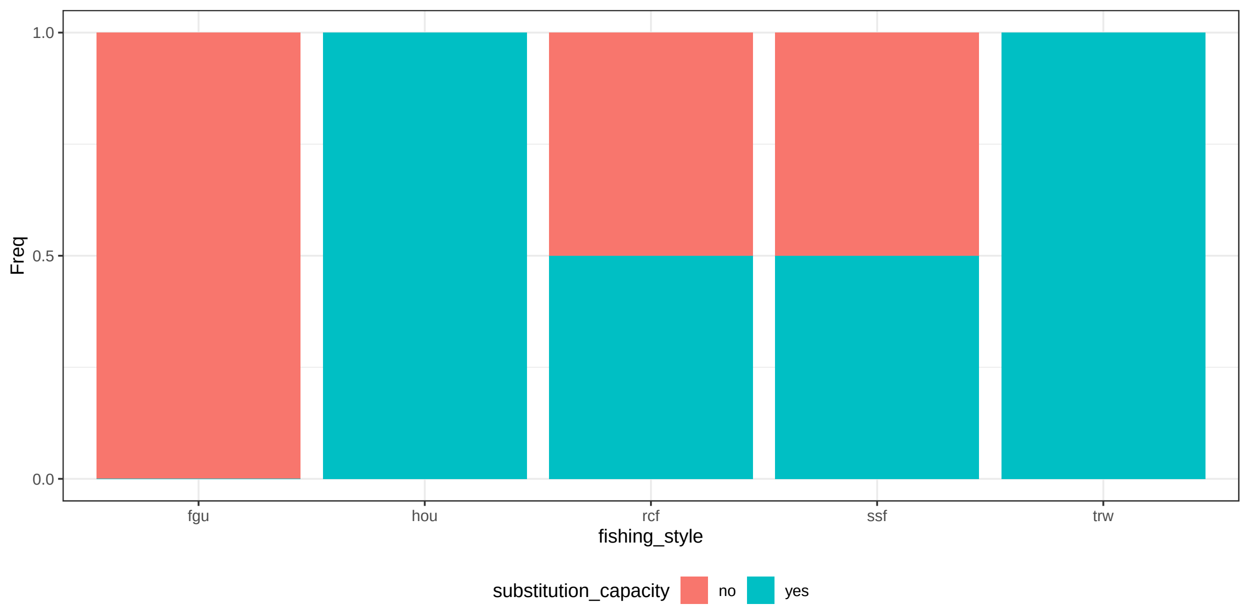


Figure C.14: conditional distribution for substitution_capacity.

## Non monetary_value

**Parents**: health, satisfaction, substitution_capacity, go_fishing.

**Levels**: low, medium, high.

**Description**: Summarizes the non-monetary outcomes associated to the fishing experience. It depends on the possibility to go out fishing and to obtain satisfying catches, while staying safe. The wmin algorithm implies that all the conditions are necessary to obtain a large value.

**Method**: Case C2 with equal weight. Algorithm is wmin.

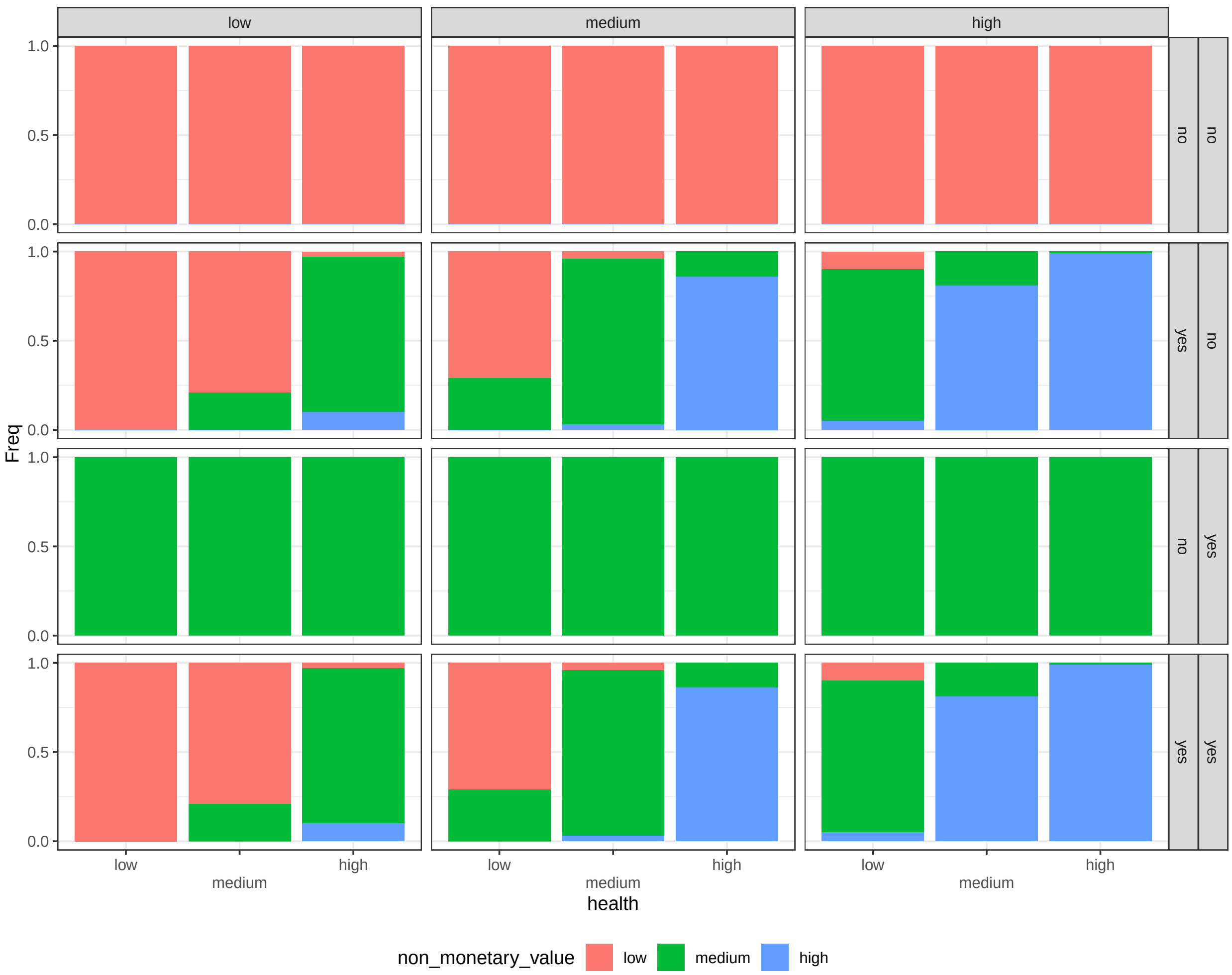


Figure C.15: conditional distribution for non_monetary_value. Columns represents levels of the node satisfaction; rows are combinations levels of the nodes substitution_capacity and go_fishing.

# Chance nodes - Economic

## Costs

**Parents**: additional_mitigation, damage, fishing_style.

**Levels**: no, low, medium, high.

**Description**: Represent how large is the costs associated with an extreme weather event compared to the annual expected costs. No is < 0.5%; Low is between 0.5% and 2.5%; Medium is between 2.5% and 5%; high is larger than 5%.

**Method**: Case C4. The CPT reflects literature and interviews as described in Annex C..

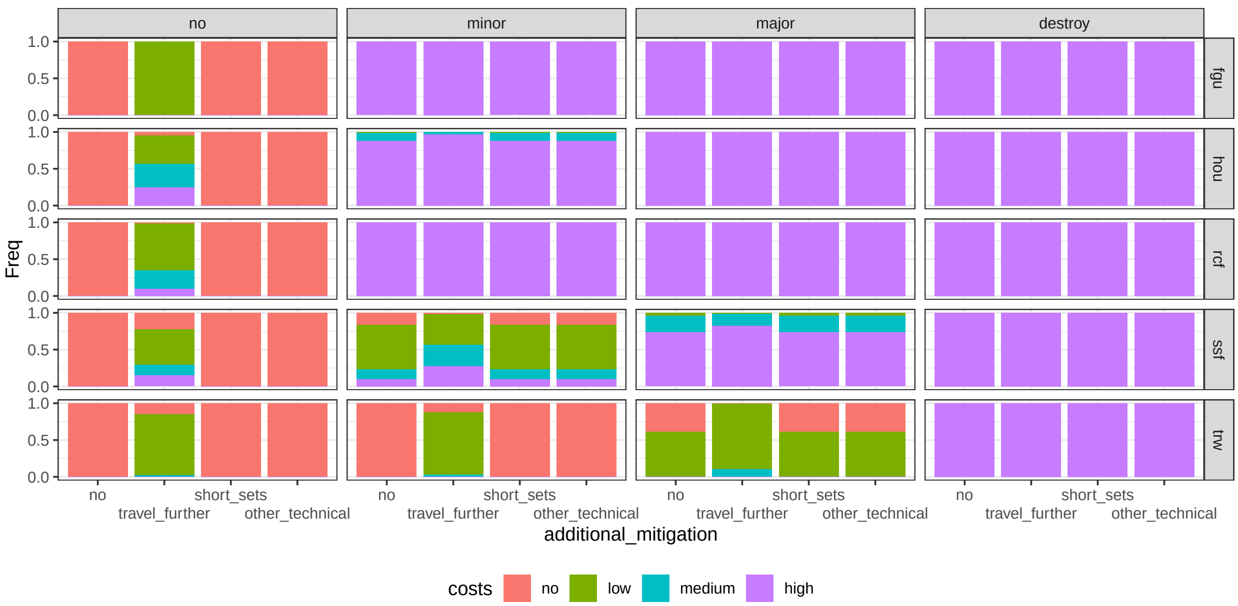


Figure C.16: conditional distribution for costs. Columns represents levels of the node damage; rows are levels of the node fishing_style.

## Monetary loss

**Parents**: substitution_capacity, costs, fishing_style, go_fishing.

**Levels**: no, low, medium, high.

**Description**: NA.

**Method**: Case C3. In the case of recreational fishers, if the node go_out is no then the cost is no. For commercial fishers, if the node go_out is no the cost depend on the estimated cost of a missed fishing day balanced on objectives type. When the node go out is yes, loss equal costs.


Figure C.17: conditional distribution for monetary_loss. Columns represents levels of the node costs; rows are combinations levels of the nodes fishing_style and go_fishing.

## Credit

**Parents**: fishing_style.

**Levels**: difficult, neutral, easy.

**Description**: The possibility to obtain loans for investments.

**Method**: Case C1 based on question Q 25.

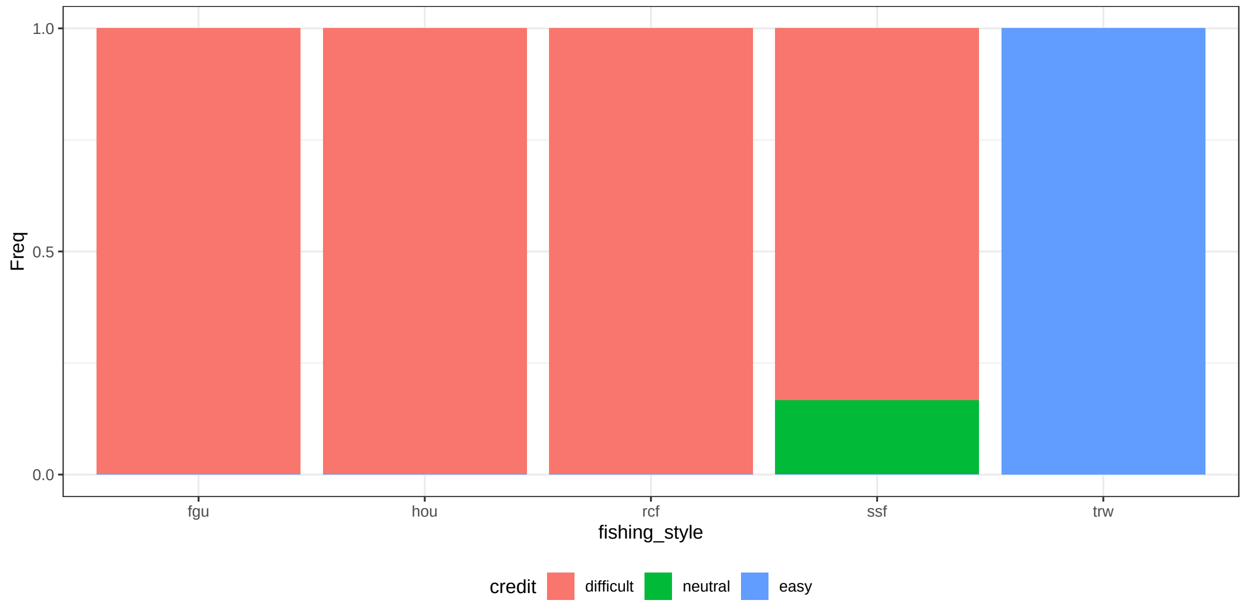


Figure C.18: conditional distribution for credit.

## Job mobility

**Parents**: fishing_style.

**Levels**: low, medium, high.

**Description**: The extent to which non-fishing income-generating activities are accessible to an individual or household.

**Method**: Case C1 based on question Q 23.

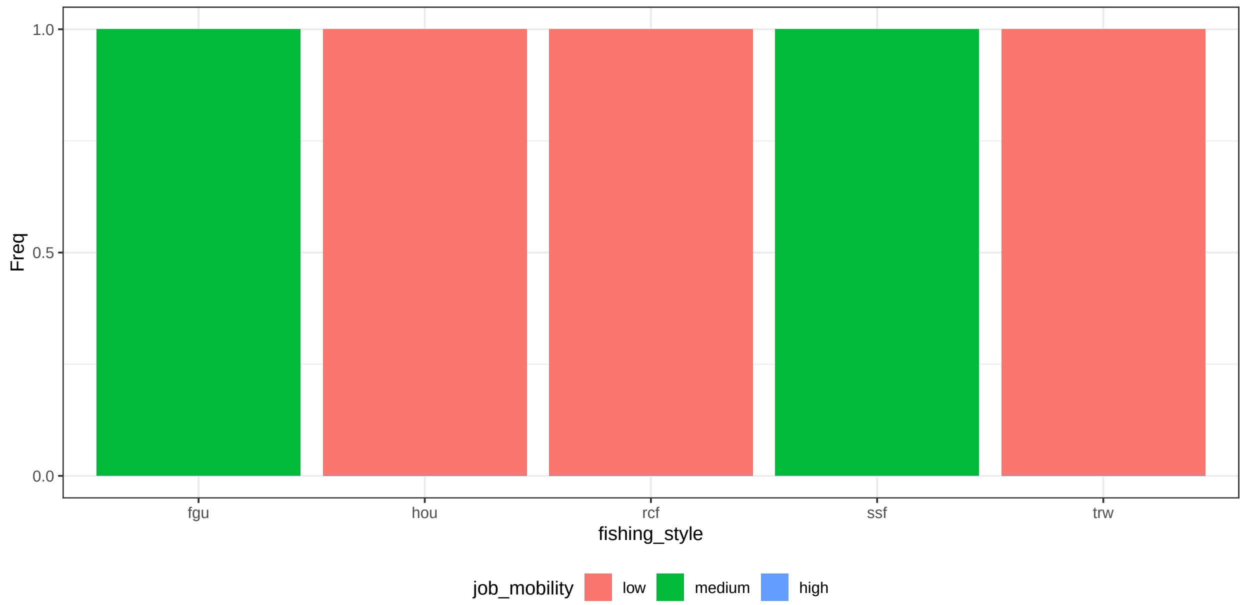


Figure C.19: conditional distribution for job_mobility.

## Economic buffers

**Parents**: job_mobility, credit.

**Levels**: low, medium, high.

**Description**: Describes the capability to buffer economic loss by obtaining credits or by replacing with other jobs. Only professionals are capable to obtain economic buffers.

**Method**: Case C2 with unequal weight, job mobility is 50% less important than credit. Algorithm is wmax.

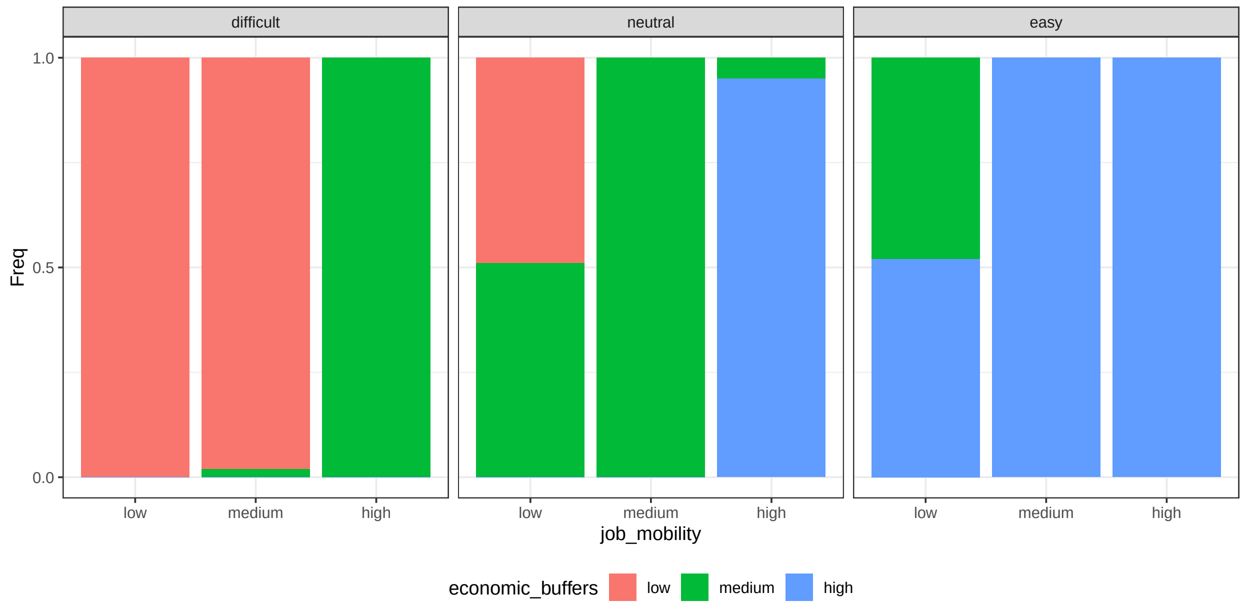


Figure C.20: conditional distribution for economic_buffers. Columns represents levels of the node credit.

Next figure represent the conditional probability distribution of the node economic buffers by fishing style

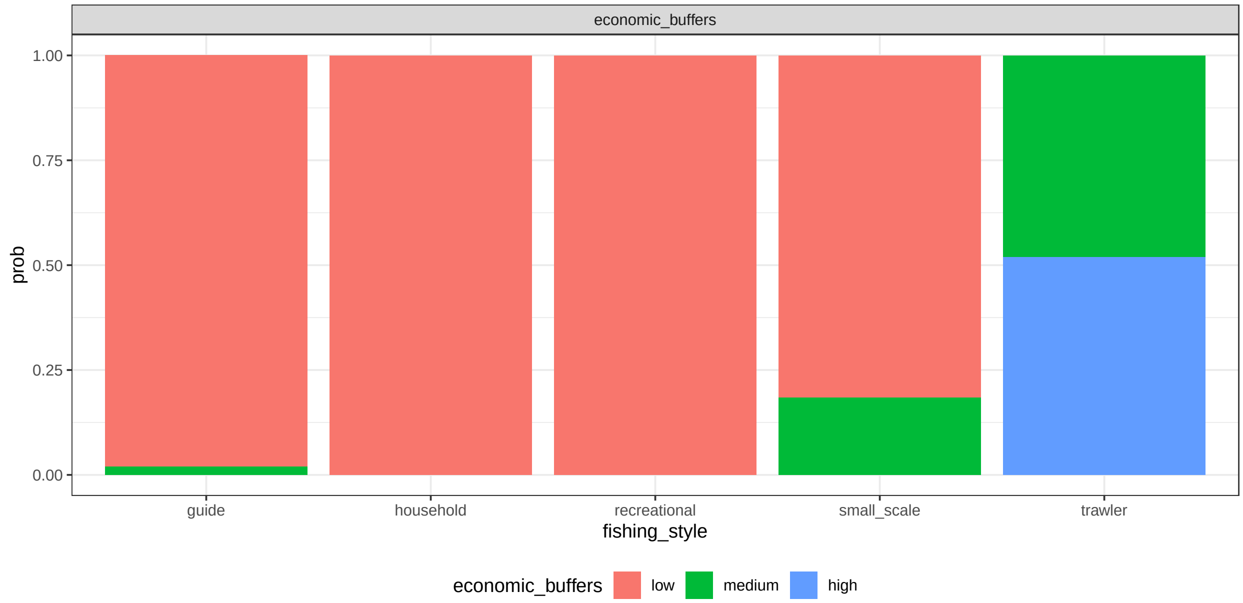


Figure C.21: conditional probability distribution of the node individual importance by fishing style

## Chance nodes - Individual

### Identity

**Parents**: fishing_style.

**Levels**: not_important, somehow_important, very_important.

**Description**: The degree to which fishing contributes to an individual's sense of identity, including self-perception.

**Method**: Case C1 based on question Q 33.

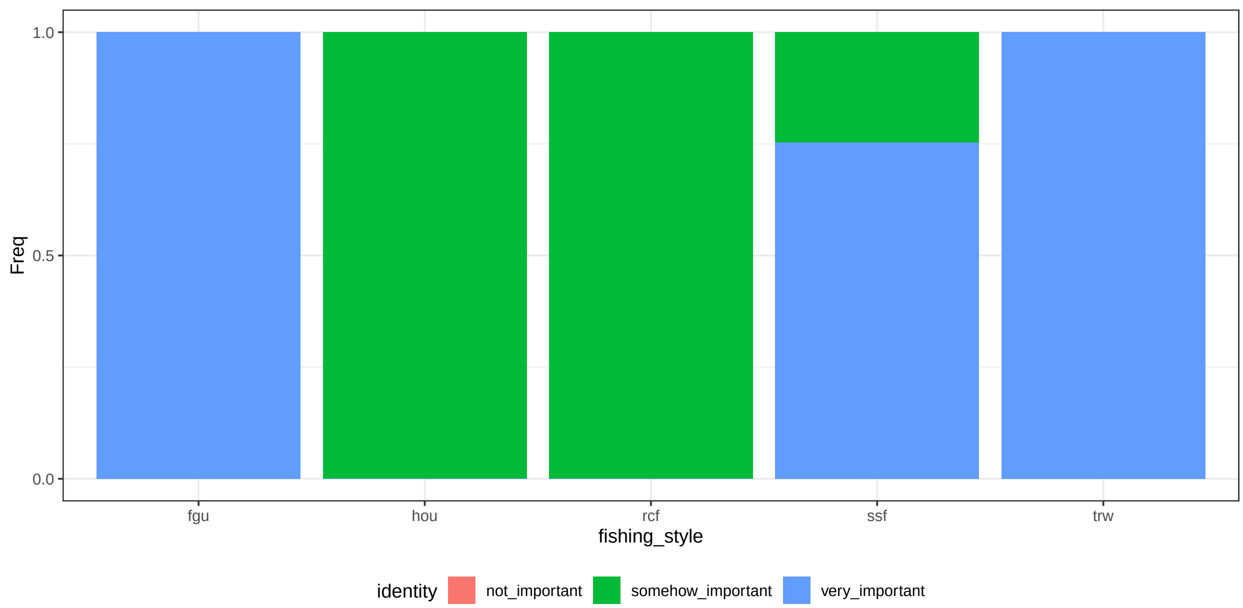


Figure C.22: conditional distribution for identity.

## Sense of_home

**Parents**: fishing_style.

**Levels**: not_important, somehow_important, very_important.

**Description**: The degree to which fishing contributes to build a sense of place/home to a specific location. It can either be at sea or at land.

**Method**: Case C1 based on question Q 34.

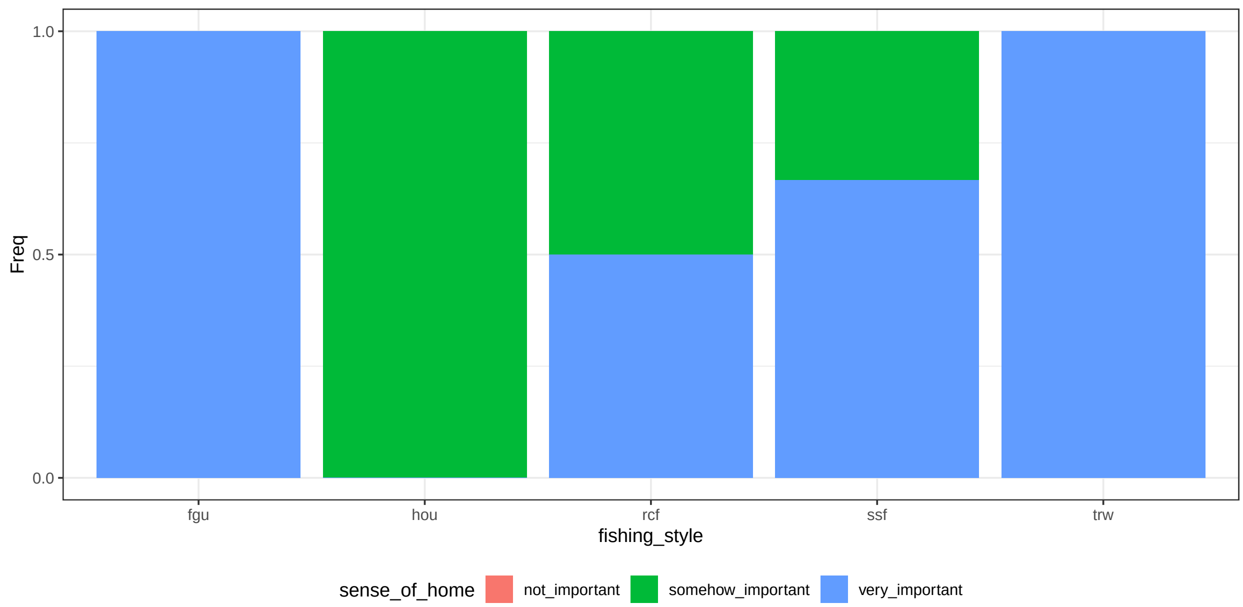


Figure C.23: conditional distribution for sense_of_home.

### Outdoor benefit

**Parents**: fishing_style.

**Levels**: not_important, somehow_important, very_important.

**Description**: The extent which an individual value the time spent outdoor (while fishing) to his personal well-being.

**Method**: Case C1 based on question Q 32.

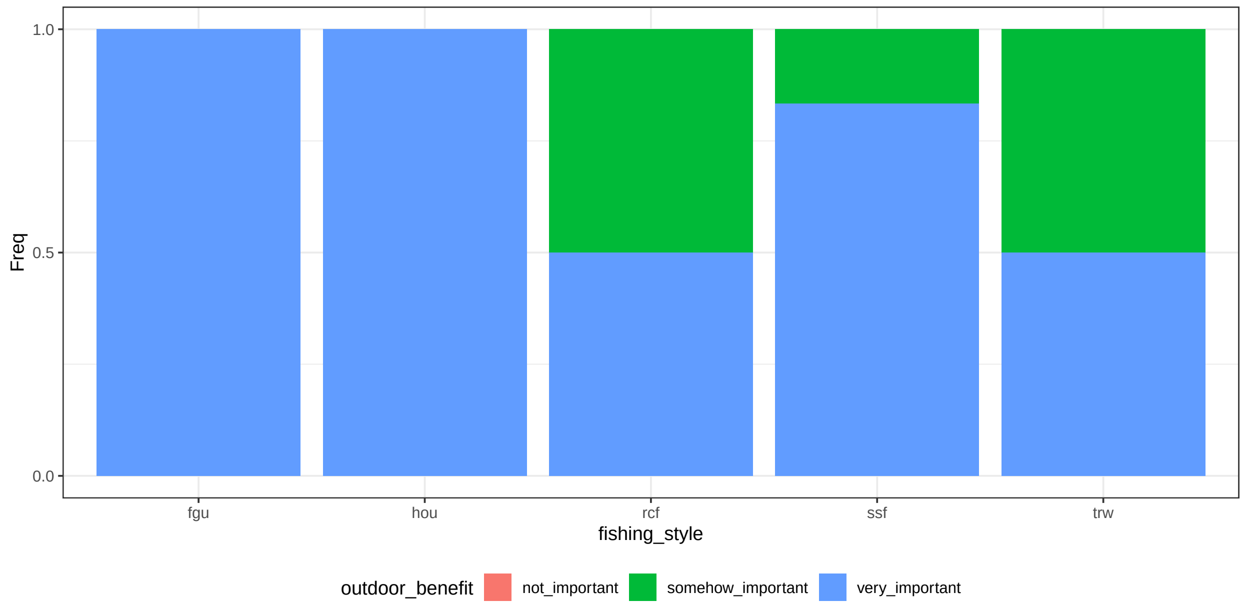


Figure C.24: conditional distribution for outdoor_benefit.

## Individual importance

**Parents**: outdoor_benefit, identity, sense_of_home.

**Levels**: low, medium, high.

**Description**: Summarizes the importance of the variables defining individual features to different fishers. The algorithm wmax gives more contribution to large states.

**Method**: Case C2 with equal weight. Algorithm is wmax.

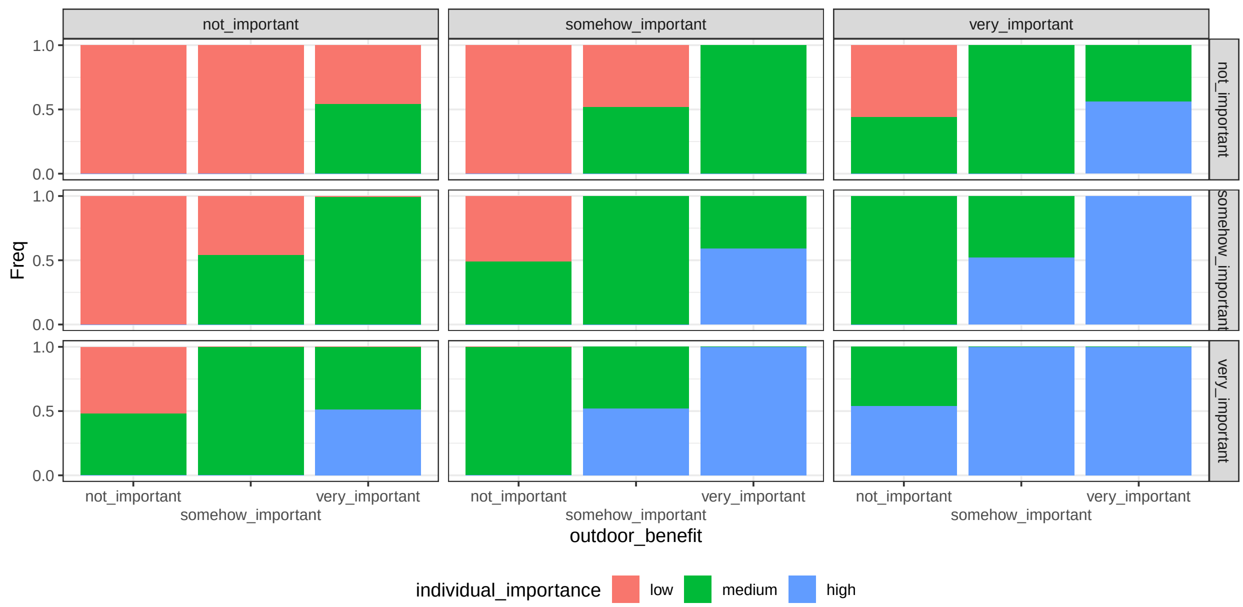


Figure C.25: conditional distribution for individual_importance. Columns represents levels of the node identity; rows are levels of the node sense_of_home.

Next figure represent the conditional probability distribution of the node individual importance by fishing style

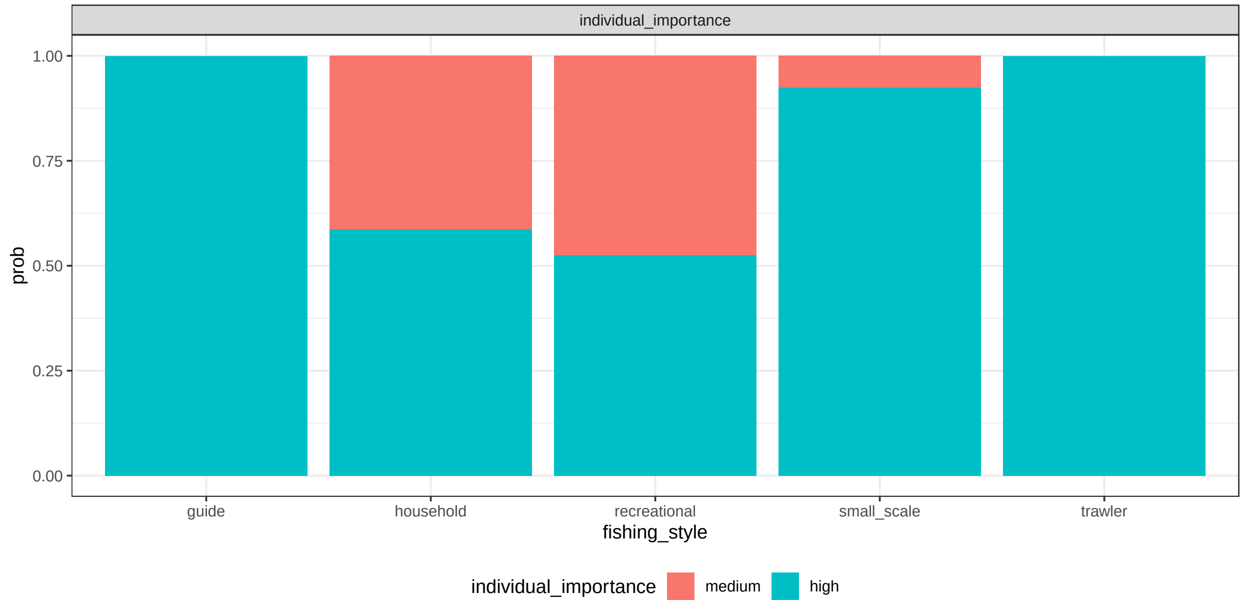


Figure C.26: conditional probability distribution of the node individual importance by fishing style

# Chance nodes - Societal

## Knowledge transmission

**Parents**: fishing_style.

**Levels**: not_important, somehow_important, very_important.

**Description**: The extent which an individual value the transmission of fishing knowledge and culture to future generations.

**Method**: Case C1 based on question Q 27.

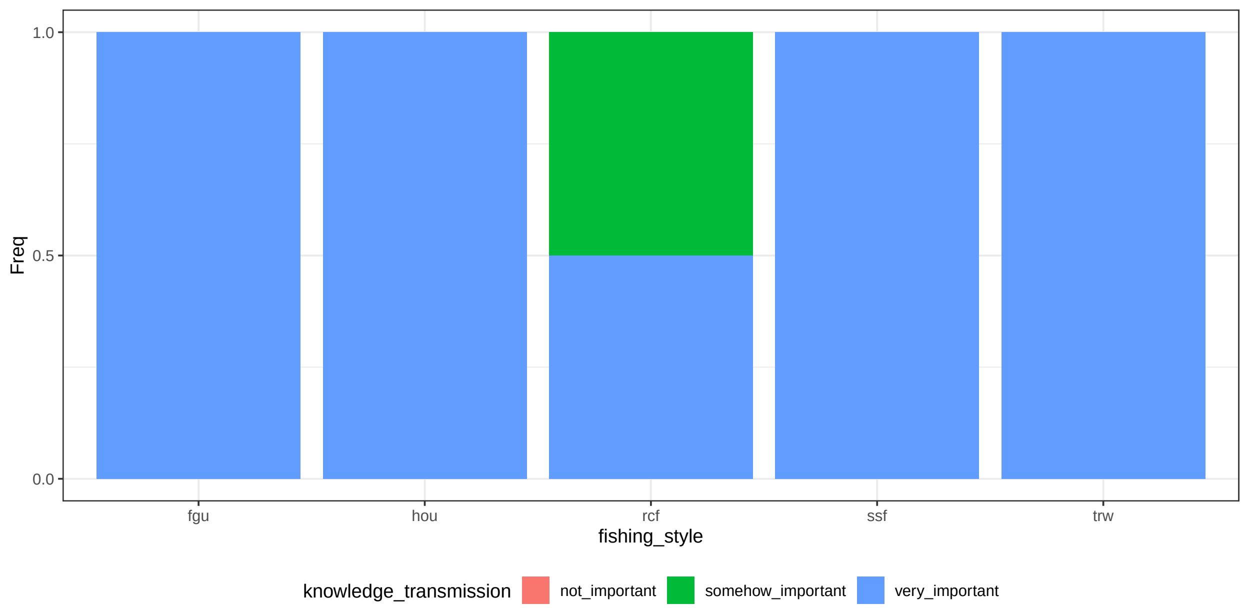


Figure C.27: conditional distribution for knowledge_transmission.

## Benefit local_community

**Parents**: fishing_style.

**Levels**: not_important, somehow_important, very_important.

**Description**: The extent which an individual value fishing to build/maintain social relations within an extended local community, such as a town or neighborhood.

**Method**: Case C1 based on question Q 28.

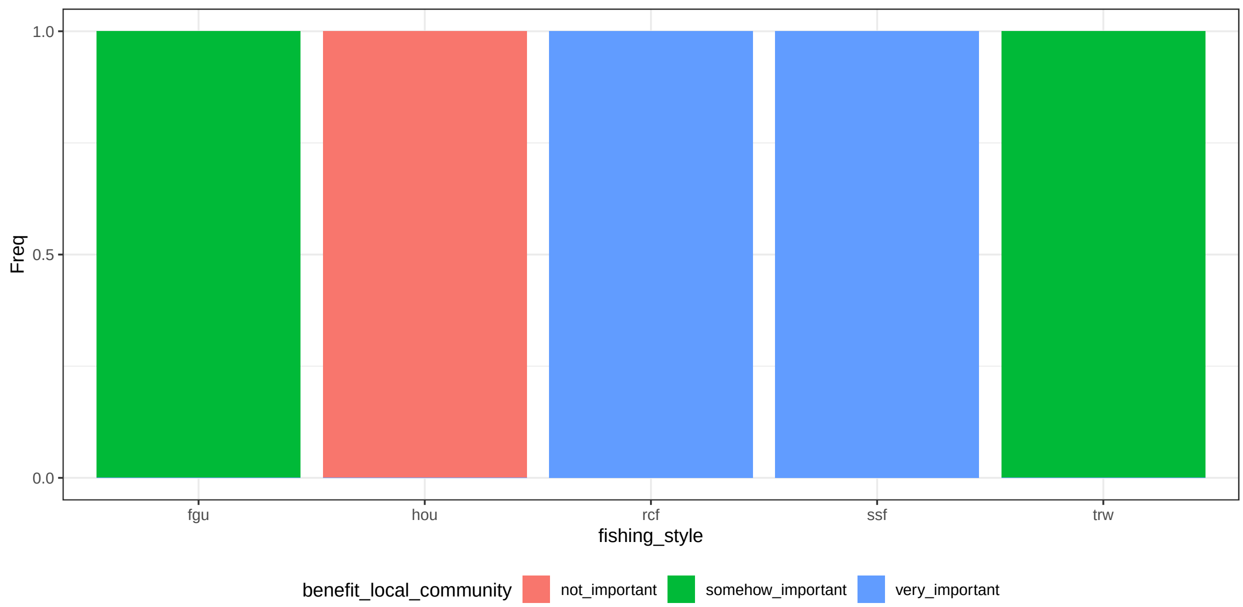


Figure C.28: conditional distribution for benefit_local_community.

### Traditional food

**Parents**: fishing_style.

**Levels**: not_important, somehow_important, very_important.

**Description**: The extent which fishing contributes to provide for traditional food.

**Method**: Case C1 based on question Q 30.

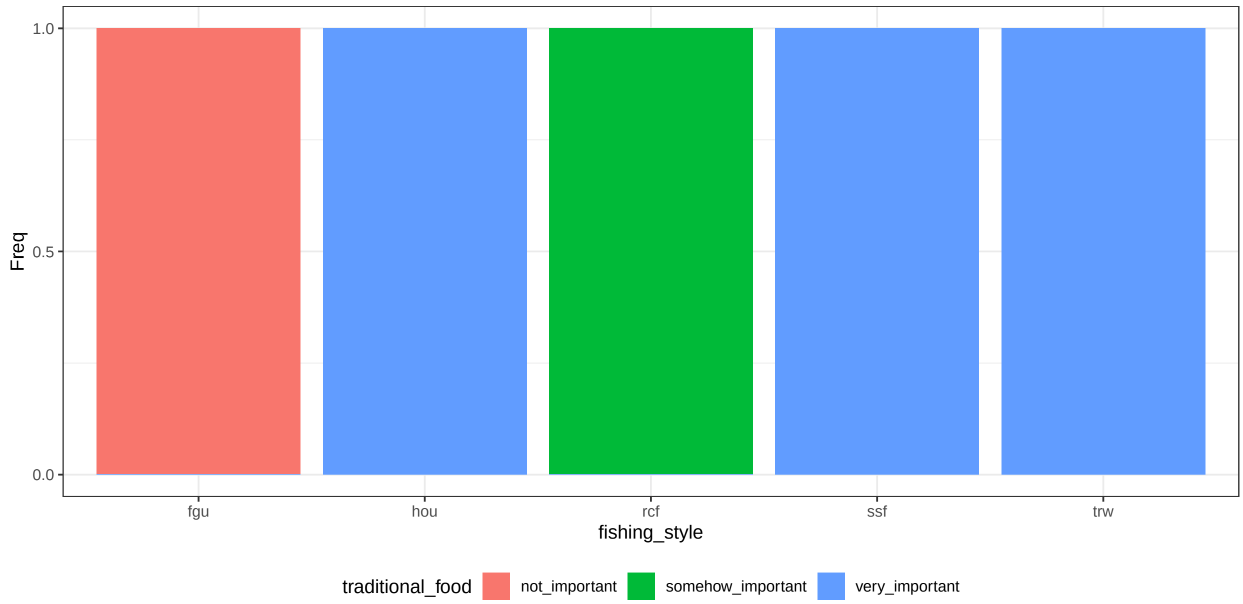


Figure C.29: conditional distribution for traditional_food.

## Benefit personal_community

**Parents**: fishing_style.

**Levels**: not_important, somehow_important, very_important.

**Description**: The extent which an individual value fishing to build/maintain social relations within a small community, such as groups of friends or associations.

**Method**: Case C1 based on question Q 29.

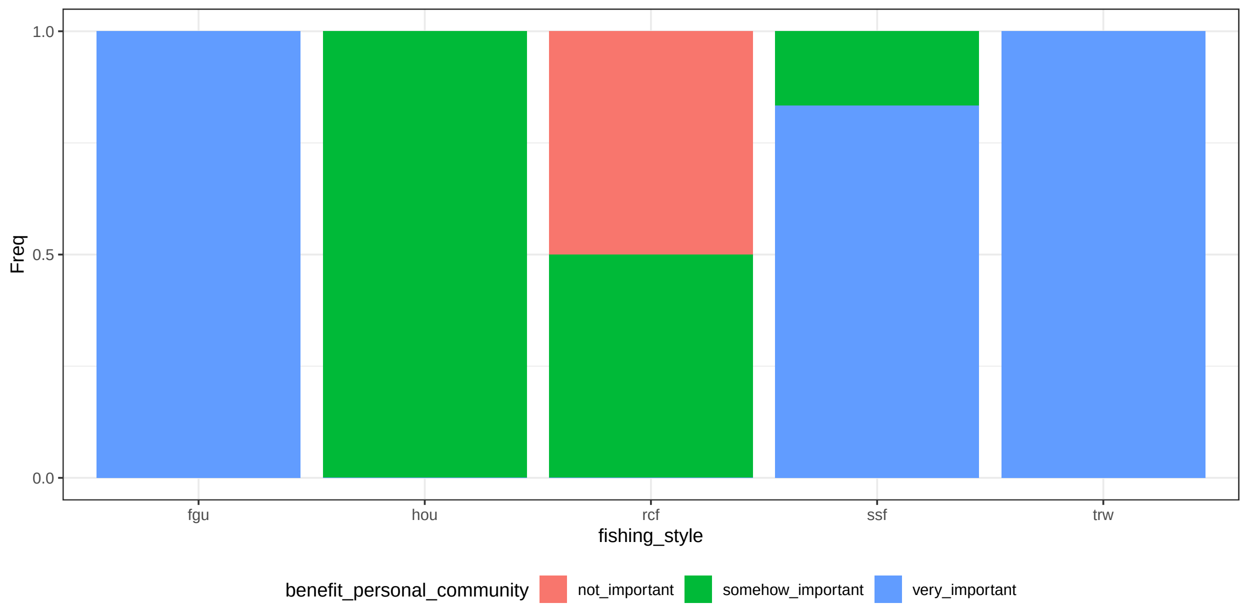


Figure C.30: conditional distribution for benefit_personal_community.

## Societal importance

**Parents**: knowledge_transmission, benefit_local_community, traditional_food, benefit_personal_community.

**Levels**: low, medium, high.

**Description**: Summarizes the importance of the variables defining societal features to different fishers. The algorithm wmax gives more contribution to large states.

**Method**: Case C2 with equal weight. Algorithm is wmax.

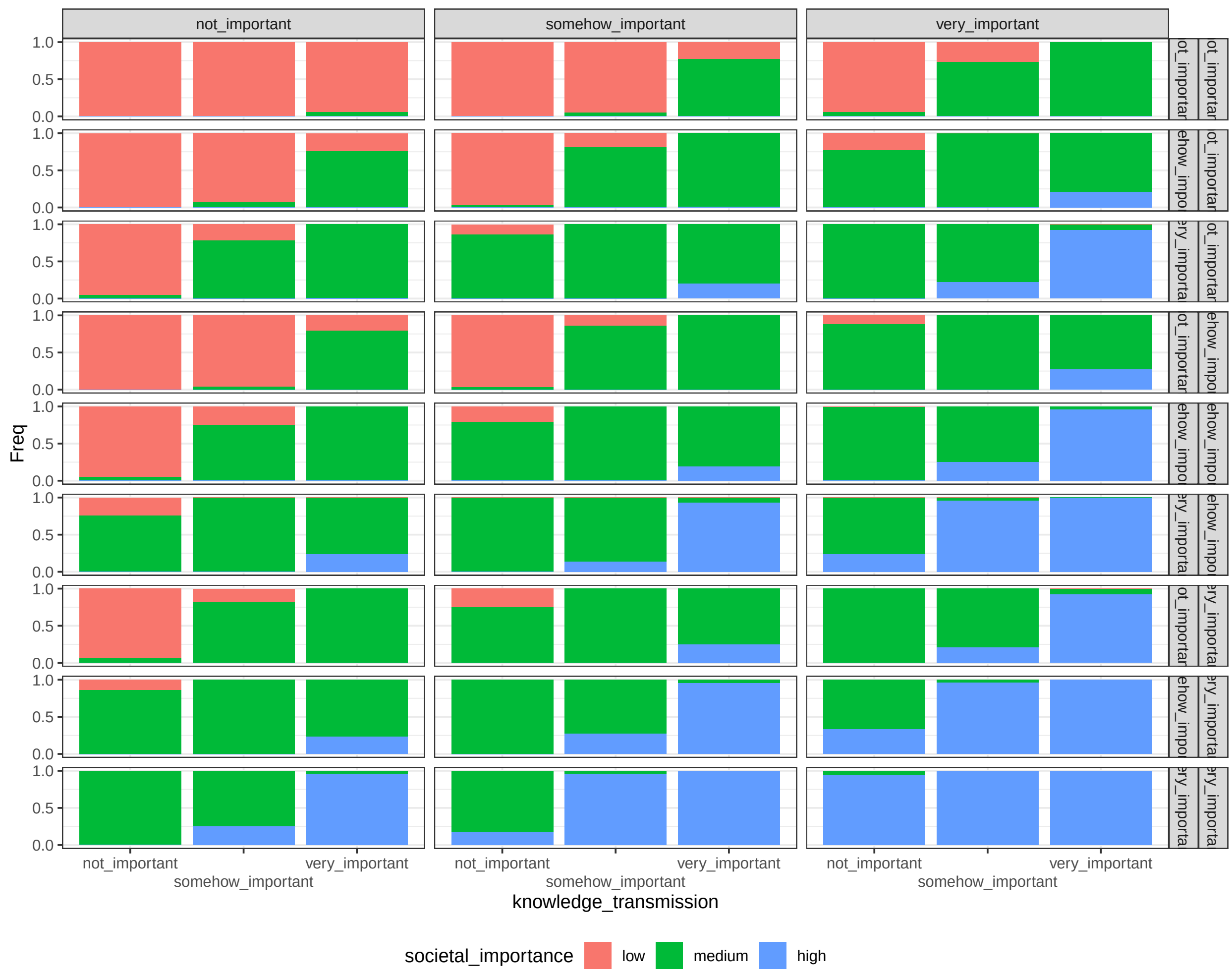


Figure C.31: conditional distribution for societal_importance. Columns represents levels of the node benefit_local_community; rows are combinations levels of the nodes traditional_food and benefit_personal_community.

Next figure represent the conditional probability distribution of the node societal importance by fishing style

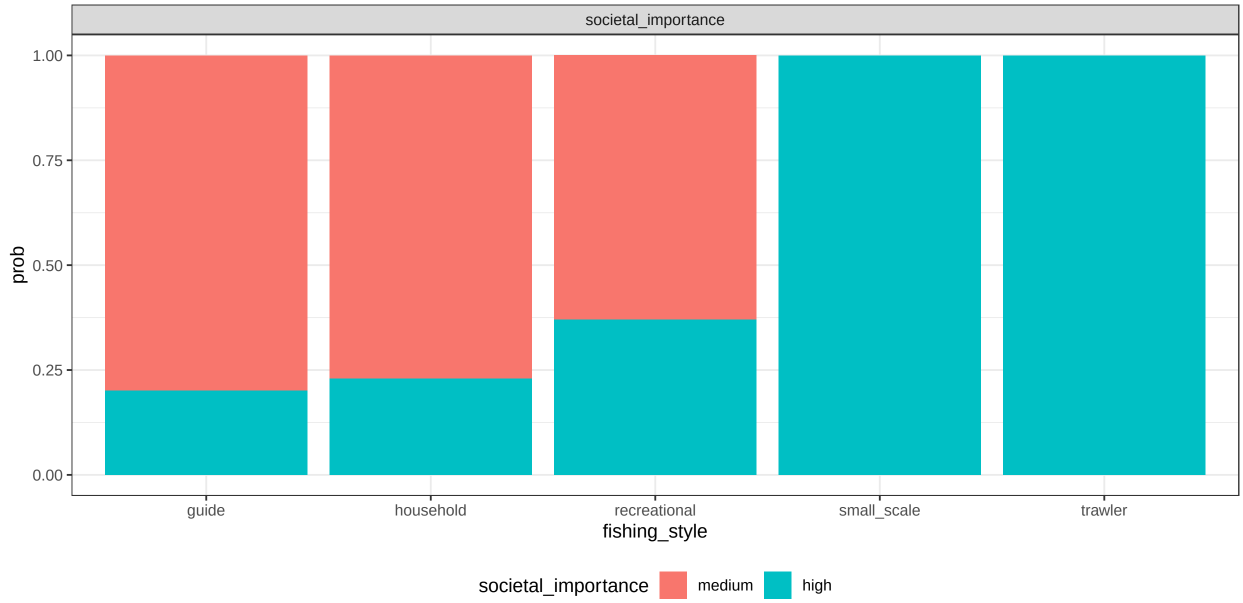


Figure C.32: conditional probability distribution of the node individual importance by fishing style

# Utilities - Risk

Shape how different fisheries will perceive risk. To each of these variables, the effect of the human community variables is to be interpreted as "how much this variable is important"? The CPT table combine the concepts of ranked nodes and the classic definition of Risk = Hazard * Vulnerability. In the case of individual and societal importance, Risk is exactly calculated as Hazard * Vulnerability. Parent categories (Low, medium and high; as well as not important, somehow important and very important) are mapped to 0, 0.5 and 1 values. The parent value are multiplied.

## Individual risk

**Parents**: individual_importance, non_monetary_value, strategy_to_change.

**Levels**: low, medium, high, not_relevant.

**Description**: The risk of having a mismatch between the outcomes that fishing provides and the importance that individual well-being receives from fishing.

**Method**: Case C5.

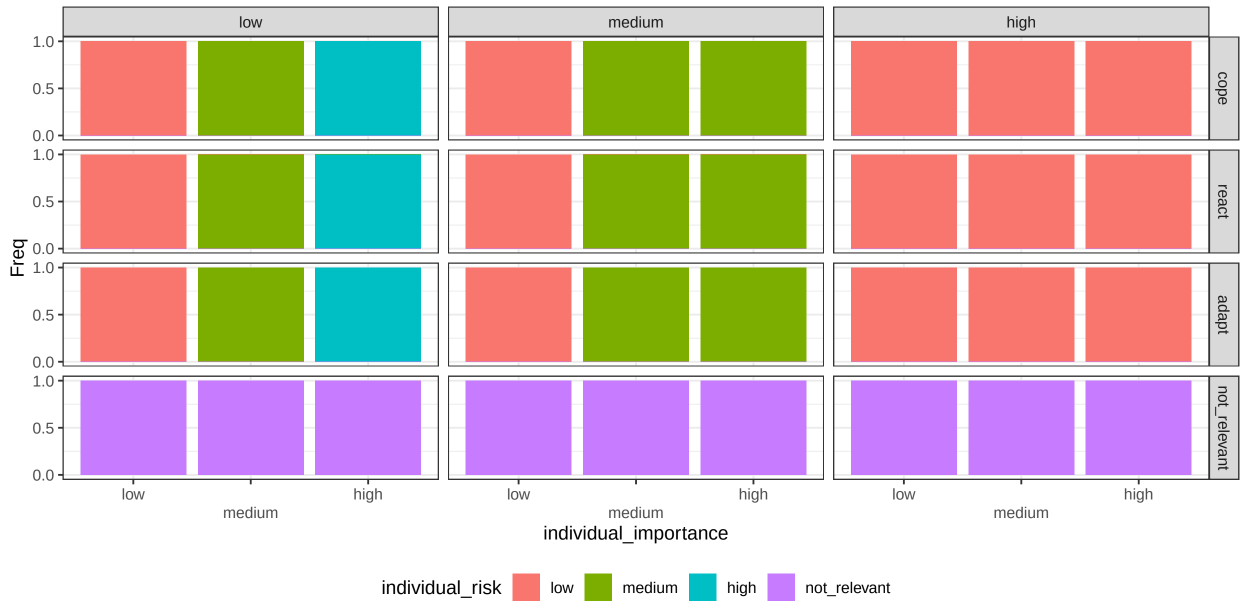


Figure C.33: conditional distribution for individual_risk. Columns represents levels of the node non_monetary_value; rows are levels of the node strategy_to_change.

## Societal risk

**Parents**: societal_importance, non_monetary_value, strategy_to_change.

**Levels**: low, medium, high, not_relevant.

**Description**: The risk of having a mismatch between the outcomes that fishing provides and the importance that fishing provides to society.

**Method**: Case C5.

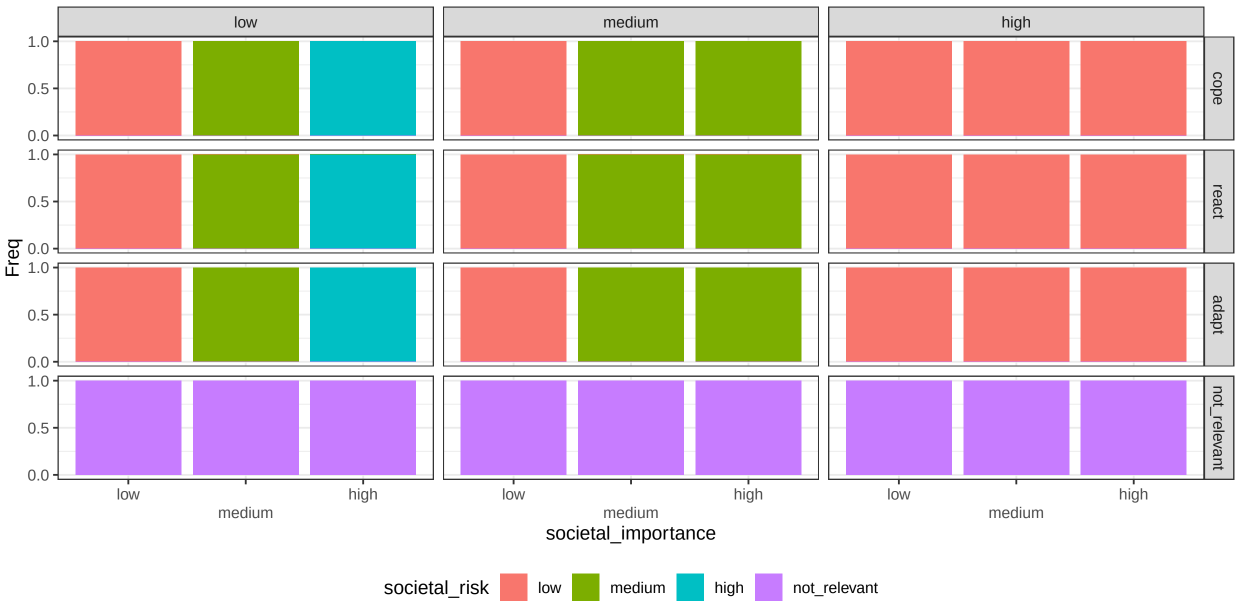


Figure C.34: conditional distribution for societal_risk. Columns represents levels of the node non_monetary_value; rows are levels of the node strategy_to_change.

## Economic risk

**Parents**: economic_buffers, monetary_loss, strategy_to_change.

**Levels**: low, medium, high, not_relevant.

**Description**: The risk of having economic fluctuations balanced for how economy is considered an important value.

**Method**: Case C2. Weight ratio for the parent *importance* and *cost* is 1:2. Algorithm is wmax.

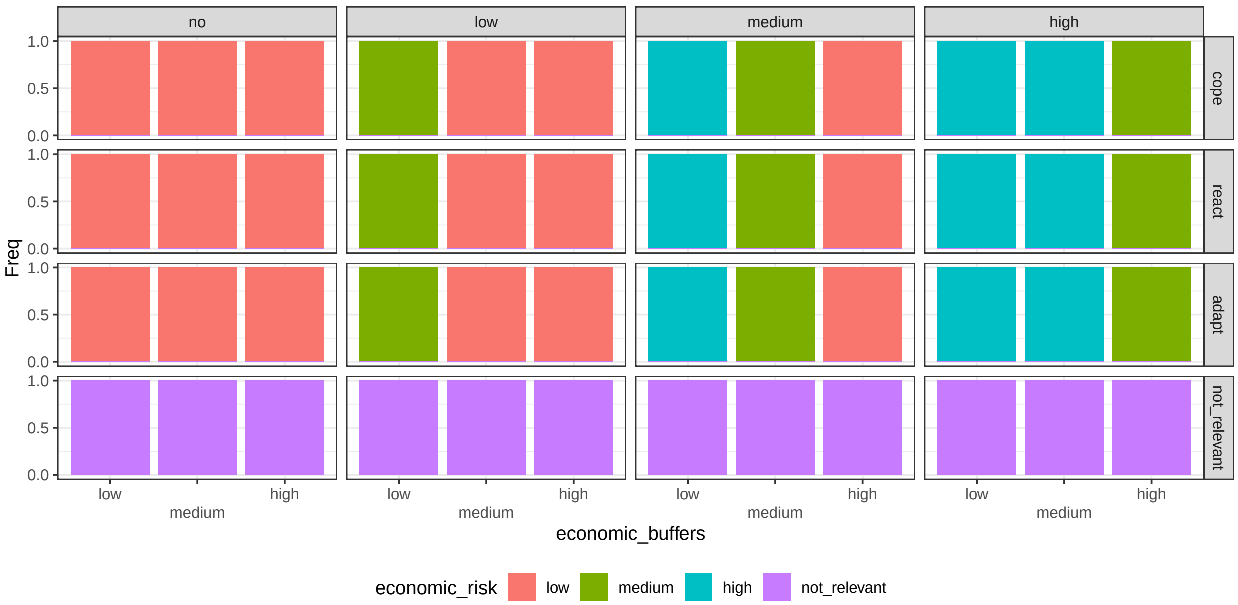


Figure C.35: conditional distribution for economic_risk. Columns represents levels of the node monetary_loss; rows are levels of the node strategy_to_change.

# References


Fenton, N. E., M. Neil, and J. G. Caballero. 2007. "Using Ranked Nodes to Model Qualitative Judgments in Bayesian Networks." *IEEE Transactions on Knowledge and Data Engineering* 19 (10): 1420–32. https://doi.org/10.1109/TKDE.2007.1073.

Fenton, Norman E., and Martin Neil. 2019. *Risk Assessment and Decision Analysis with Bayesian Networks.* Second edition. Boca Raton: CRC Press, Taylor & Francis Group.